

Dataset Overview

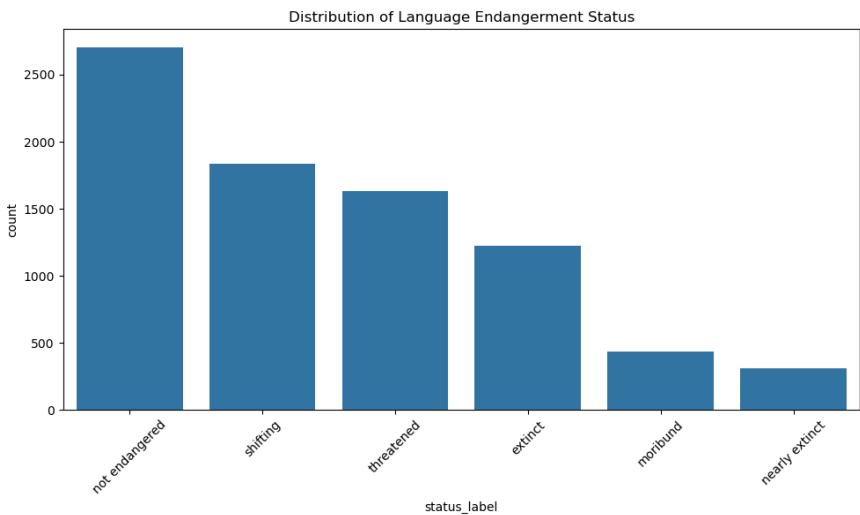
The dataset contains 8,612 records and 12 columns documenting languages globally, focusing on their geographic coordinates, linguistic lineage, and endangerment status. Key variables include macroarea, latitude, longitude, and whether the language is an isolate or part of a broader family.

Data Quality Notes

- Target variables `status_code` and `status_label` have approximately 5.5% missing values, which will need to be addressed prior to predictive modeling.
- Geographic and spatial features contain minor gaps: `latitude` and `longitude` are missing 3.6%, `macroarea` 2.6%, and `countries` 1.2%.
- Linguistic taxonomy identifiers show varying completeness, with `iso639p3code` missing 8.8% and `family` missing 2.1%.

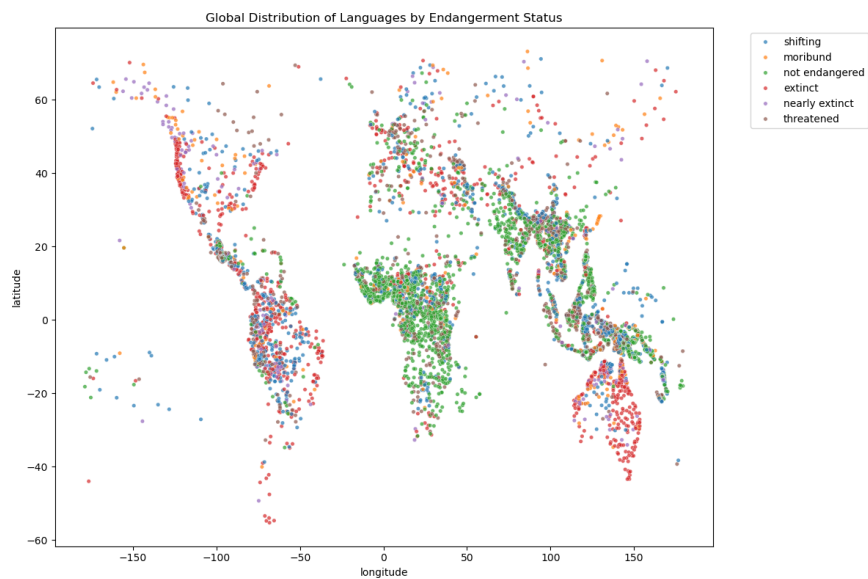
Exploratory Analysis

Chart 1: Distribution of Language Endangerment Status



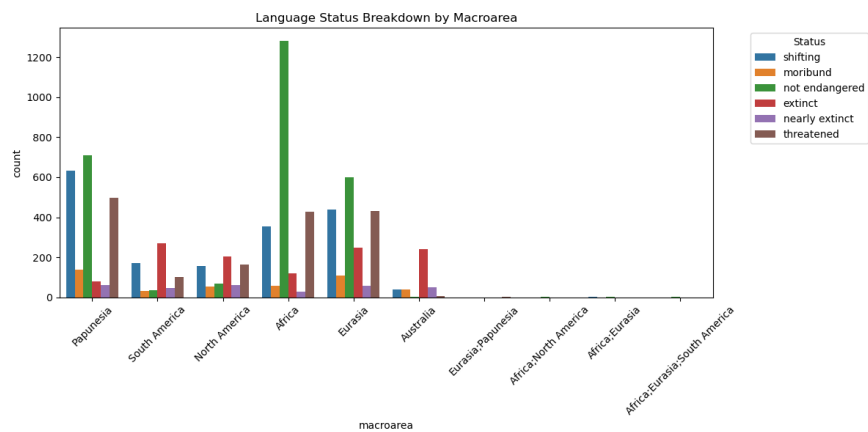
What it shows: The majority of languages are classified as not endangered, showing significant class imbalance.

Chart 2: Global Distribution of Languages by Endangerment Status



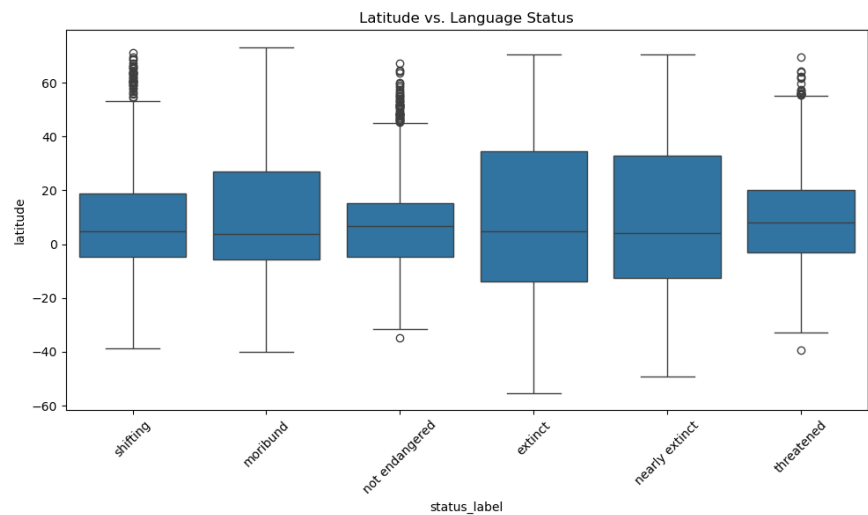
What it shows: Language endangerment is globally distributed, but with noticeable clusters in certain geographic regions.

Chart 3: Language Status Breakdown by Macroarea



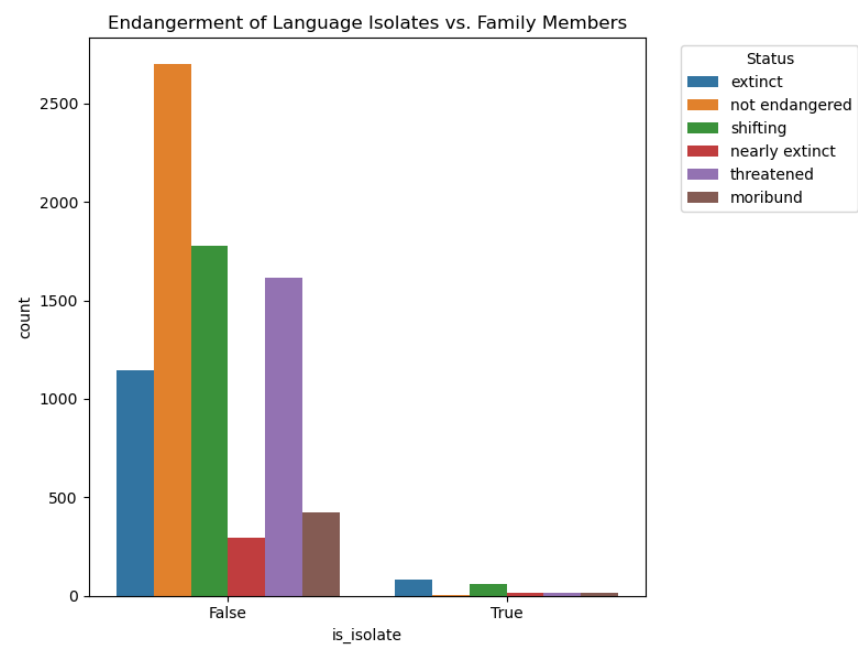
What it shows: Regions like Papunesia have large numbers of not endangered languages, whereas others have higher endangerment proportions.

Chart 4: Latitude vs. Language Status



What it shows: Endangered languages appear across varied latitudes, with slight differences in median latitude among statuses.

Chart 5: Endangerment of Language Isolates vs. Family Members



What it shows: Language isolates are much rarer overall but exhibit varied distributions of endangerment statuses.

Modeling Results

Class	Precision	Recall	F1-Score	Support
extinct	0.61	0.65	0.63	245
moribund	0.05	0.02	0.03	87
nearly extinct	0.20	0.13	0.16	62
not endangered	0.61	0.72	0.66	541
shifting	0.45	0.46	0.45	367

Class	Precision	Recall	F1-Score	Support
threatened	0.40	0.35	0.37	326
Accuracy			0.51	1628
Macro Avg	0.39	0.39	0.38	1628
Weighted Avg	0.49	0.51	0.50	1628

Key Insights

The baseline Random Forest model achieved a moderate accuracy of 51.41%. It performed adequately on majority classes like "not endangered" and "extinct" but struggled heavily with minority categories, evidenced by an F1-score of just 0.03 for "moribund" languages. The vast majority of the dataset consists of "not endangered" languages (over 2,700), establishing a strong baseline of language vitality but causing severe class imbalance.

Geographic and spatial features are critical indicators, as vitality and endangerment cluster visibly across macroareas. For instance, Africa and Papunesia harbor robust populations of vital languages, whereas other regions lean heavily toward shifting or extinct status. The dataset currently lacks dynamic historical or socioeconomic context, limiting predictive capabilities primarily to geographic boundaries and basic linguistic lineage (e.g., family vs. isolates).

Recommendations / Next Steps

- Apply data resampling techniques (like SMOTE) or group minor endangerment categories to handle the severe class imbalance limiting model performance on rare classes.
- Augment the dataset with external socio-economic, historical, or policy data on a per-country basis to provide the model with stronger predictors than just spatial coordinates.
- Investigate handling of missing target variables (5.5%) by either careful imputation using linguistic family tendencies or filtering them from the modeling pipeline.
- Explore spatial or clustering algorithms (e.g., K-Means on latitude/longitude) to create geographic interaction features that might better predict language shift.